MUHAMMAD RIFKY NAUFAL

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SUMMARY

Fifth-semester Applied Bachelor's (D4) student in Electrical Automation Engineering at Institut Teknologi Sepuluh Nopember (ITS), GPA 3.50/4.00. Won 1st place at the National Selection of WorldSkills ASEAN 2025 (Mechatronics) and 2nd place at the 2023 National Vocational Skills Competition (LKS-SMK). Since 2026, has been building IoT-based disaster early warning prototypes at PUSDI MKPI ITS and assisting practical sessions at the PLC & Supervisory Control System Laboratory. Seeking an internship in automation, electrical, and instrumentation.

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS), Surabaya 2024 - Present
Applied Bachelor's Degree (D4), Electrical Automation Engineering | Semester 5 | GPA 3.50/4.00

SMK Negeri 1 Jenangan, Ponorogo Jun 2020 - May 2024
Industrial Automation Engineering (4-year vocational program)

AWARDS

- **Finalist, Design Thinking Competition,** Department of Electrical Automation Engineering, ITS (Jul 2026)
- **1st Place, National Selection,** 14th WorldSkills ASEAN Competition 2025, Mechatronics, Ministry of Manpower, BBPVP Serang (Nov 2024). Top 5 teams from regional selections.
- **3rd Place, Regional Selection,** 14th WorldSkills ASEAN Competition 2025, Mechatronics, BPVP Sidoarjo (Oct 2024)
- **2nd Place, National Level,** Vocational Students Competition (LKS-SMK), Mechatronics, Puspresnas, Ministry of Education (Oct 2023)
- **1st Place, Provincial Level,** 31st LKS-SMK East Java, Mechatronics, East Java Provincial Education Office (May 2023)

EXPERIENCE

ITS Professional Certification Body (LSP ITS) Aug 2026 - Present
Assessor Assistant

- Verify the safety of electrical circuits built by about 15 candidates per competency test session, before the circuits are powered.

PUSDI MKPI ITS (Center for Disaster Mitigation and Climate Change Studies) Feb 2026 - Present
Researcher, Disaster Mitigation and Climate Change

- Design and test embedded system prototypes for IoT-based disaster early warning.
- **INSAMO FLOWS (flood monitoring):** Built a web-based system on an ESP32-S3. An A01NYUB ultrasonic sensor measures water level, a DS18B20 sensor measures temperature, and data is sent over RS485.
- **INSAMO SIGMA (earthquake mitigation):** Developed a prototype with sensor modules, embedded communication, and a web-based geolocation dashboard.
- **Low-cost weather station:** Designed a weather station with integrated sensors on a microcontroller that logs measurement data.

PLC & Supervisory Control System Laboratory, ITS Jan 2026 - Present
Laboratory Assistant

- Guide practical sessions on Electrical Measurement Circuits, Electrical Machines, and Digital Circuits for 9 students (5 from the 2026 cohort and 4 from the 2025 cohort).
- Maintain laboratory equipment so it is ready for practical sessions.

STP Automotive ITS - Electric Go-Kart Team Oct 2025 - Mar 2026
Member, Battery and Electrical Systems Division

- Designed the wiring schematic and integrated high-power components into the go-kart drivetrain.
- Designed and tested Battery Management System (BMS) wiring, and traced electrical faults.

ABU Robocon ITS Sep 2024 - Oct 2024
Intern, Electrical Division

- Developed the electrical control system for a competition robot.
- Installed wiring, integrated sensors and actuators, and tested signals in the robot control system.

PT Cipta Otomasi Indonesia Jan 2024 - Apr 2024

Vocational School Internship, Automation Engineering

- Assembled and wired pneumatic and electro-pneumatic trainers on a Modular Production System (MPS).
- Connected industrial PLCs to pneumatic actuators and control panels, then tested the system and traced faults.

Workshop Proteksi Sep 2023 - Dec 2023

Vocational School Internship, Industrial Engineering

- Built and assembled automatic gates, automatic sliding doors, a conveyor for moving goods, and a food elevator.
- Wired control panels, and did mechanical fabrication, welding, and structural assembly.

TRAINING & TEACHING

Trainer, Micro Line Follower and IoT Training

- Trained about 20 participants per session on Micro Line Follower robots (Arduino and ESP). Our team designed the microcontroller board (PCB) and the acrylic body, cut the acrylic, and assembled the units. Participants used the finished boards.
- Taught IoT at basic, intermediate, and advanced levels using microcontroller boards designed by our team.
- Participants included vocational and senior high school students, vocational school teachers, recent graduates already working, and a robotics tutor.

Trainer, Basic Arduino Training, Surabaya City Education Office Dec 2025

- Trained about 30 participants.

Speaker, Basic and Advanced PLC Training Feb 2026

- Taught PLC programming on existing trainer kits, with about 20 participants per session.

Trainer, ITS Innovation Experience: Smart Robot for Beginner May 2026

PROJECTS

Oil Skimmer Boat (Oil Spill Collector), Freelance Electrical Lead Feb 2026 - Jun 2026

- Delivered the boat's electrical system for a student team: designed the system, wired the components, and integrated the power system with the boat's propulsion and controls.

Mini Conveyor with PID Control Aug 2025 - Jan 2026

- Built a conveyor speed control system with a PID algorithm on an ESP32.
- Drove a DC motor directly and an AC motor through an inverter.

Electronics Trainer Modules (Semester 1 and 2 Project) Aug 2024 - Jun 2025

- Designed and assembled 6 trainer modules: power supply, current and voltage divider, resistor-capacitor circuit, DC motor, 2-bit adder, and 2-digit counter.

TECHNICAL SKILLS

PLC: Siemens TIA Portal (Ladder, GRAPH/SFC), Omron CX-Programmer, Mitsubishi GX Works2, Schneider Machine Expert Basic, Haiwell PLC

HMI: Weintek EasyBuilder Pro, Omron CX-Designer, Schneider Vijeo Designer, Haiwell HMI

Electrical and control: panel wiring and control schematics, sensor and actuator integration, pneumatics and electro-pneumatics, DC and AC motors (inverter), interlocking

Control algorithms: PID, on-off, sequential control

Embedded and IoT: ESP32, Arduino, UART, RS485, Blynk

Electronics design: KiCad, EasyEDA, PCB layout

CAD and manufacturing: Autodesk Inventor, Fusion 360, 3D modeling, 3D printing

ORGANIZATIONS

- **Training Division Expert Staff,** IARC XII (2026 - 2027)
- **Deputy Manager,** SENERGY ITS Student Activity Unit (2026)
- **Equipment Division Staff,** Thematic Community Service (KKN), PUSDI MKPI ITS (2026)